

MATAM, Senegal

WMO: 616300

Lat: 15.630N

Lon: 13.250W

Elev: 15

StdP: 101.14

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	17.1	-1.3	3.4	25.9	0.2	3.8	24.9	6.9	26.4	5.9	26.5	0.7	40	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.7	44.3	24.6	43.3	24.3	42.2	24.0	28.7	35.7	28.2	35.0	27.9	34.5	1.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	32.2	26.7	22.3	31.8	26.2	21.7	31.4	93.1	35.6	90.8	35.1	89.0	34.6	36.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.4	4.4	3.7	12.6	46.4	1.3	0.7	11.7	46.9	10.9	47.3	10.1	47.7	9.1	48.1		
			7.7	30.6	1.0	1.9	7.0	31.9	6.5	33.1	5.9	34.1	5.2	35.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.0	25.4	27.8	31.3	34.2	35.9	35.0	32.0	30.3	30.6	32.4	30.1	26.6
	DBStd	3.71	2.16	2.49	2.48	1.94	1.81	2.18	2.07	1.87	1.80	1.66	1.75	2.22
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7654	478	499	659	725	802	749	681	630	617	696	603	514
	CDD18.3	4612	220	266	401	475	544	499	422	372	367	438	353	256
	CDH23.3	66595	2344	3214	5758	7682	9259	8298	6337	5038	4982	6367	4559	2757
	CDH26.7	42456	1207	1849	3714	5372	6793	5921	3929	2711	2724	3977	2757	1501
Wind	WSAvg	1.5	1.3	1.5	1.4	1.5	1.8	2.2	2.0	1.6	1.4	1.1	0.9	1.2
Precipitation	PrecAvg	373	0	1	0	0	1	28	91	137	97	18	1	2
	PrecMax	704	4	22	2	4	23	108	219	315	297	117	46	42
	PrecMin	166	0	0	0	0	0	0	11	31	28	0	0	0
	PrecStd	126	1	4	0	1	4	27	51	60	55	26	6	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	40.6	43.2	44.9	45.9	45.1	41.0	39.0	39.0	41.9	40.8	38.8
		MCWB	20.5	22.1	22.8	24.4	24.3	24.9	25.3	26.5	27.5	25.2	22.8	21.9
	2%	DB	35.6	38.6	42.0	43.7	44.7	43.5	39.3	37.1	37.3	40.5	39.6	37.1
		MCWB	19.5	20.6	22.7	24.2	24.8	25.0	26.0	26.9	27.6	25.4	22.7	20.7
	5%	DB	34.2	37.2	40.8	42.7	43.8	42.1	38.0	35.9	36.2	39.4	38.5	35.6
		MCWB	18.8	20.0	22.2	23.7	24.5	25.3	26.0	27.1	27.6	25.4	22.3	20.3
	10%	DB	32.9	35.6	39.2	41.5	42.5	40.8	36.9	34.8	35.0	38.2	37.3	33.9
		MCWB	18.2	19.2	21.6	23.2	24.2	25.2	25.9	27.0	27.5	25.3	22.0	19.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	24.1	25.0	27.8	28.2	28.7	28.5	28.8	29.4	29.2	26.2	23.4
		MCDB	34.2	36.9	39.6	40.2	40.7	38.7	35.9	35.0	35.8	36.5	35.7	36.6
	2%	WB	20.8	22.3	24.0	25.9	27.0	27.8	27.8	28.2	28.7	28.3	25.1	22.0
		MCDB	33.6	35.8	39.7	41.4	40.8	38.2	35.3	34.3	34.8	36.1	35.3	35.3
	5%	WB	19.6	21.0	23.1	24.8	26.2	27.1	27.4	27.9	28.3	27.7	24.0	20.9
		MCDB	32.8	34.6	38.8	40.8	40.4	37.5	34.7	33.7	34.1	35.4	34.6	34.1
	10%	WB	18.6	20.0	22.1	24.0	25.4	26.5	27.0	27.5	27.9	27.1	23.1	19.9
		MCDB	31.6	34.0	37.5	39.7	39.4	37.0	34.1	33.0	33.4	34.5	34.7	32.6
Mean Daily Temperature Range	5% DB	MDBR	15.1	15.2	15.6	15.0	13.7	11.6	9.4	8.5	9.0	12.0	15.3	15.2
		MCDBR	17.3	17.0	17.3	16.2	15.2	13.1	10.7	9.8	10.3	13.8	16.8	17.5
		MCWBR	7.9	7.5	7.5	6.9	6.0	4.5	3.3	3.3	3.3	4.6	6.4	7.5
	5% WB	MCDBR	16.3	15.6	15.9	14.7	13.0	11.5	9.6	9.0	9.3	11.4	14.6	16.4
		MCWBR	8.0	7.6	7.6	7.0	5.7	4.4	3.5	3.5	3.9	4.4	6.2	7.6
Clear-Sky Solar Irradiance	taub	0.535	0.562	0.638	0.660	0.712	0.777	0.741	0.652	0.619	0.606	0.515	0.507	
	taud	1.789	1.733	1.592	1.597	1.509	1.467	1.559	1.733	1.788	1.784	1.907	1.890	
	Ebn at Noon	752	756	714	699	654	605	630	695	718	712	765	763	
	Edh at Noon	211	233	275	274	295	304	278	236	223	218	186	186	
All-Sky Solar Radiation	RadAvg	5.20	5.85	6.66	7.03	6.95	6.56	6.21	5.93	5.97	5.79	5.28	4.95	
	RadStd	0.30	0.28	0.26	0.24	0.25	0.24	0.20	0.23	0.21	0.20	0.23	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page